

Inventory System Data Flow

Overview

The inventory system uses three interconnected tables to track stock movements, costs, and current balances:

- **inventory_transactions**: Immutable ledger of every transaction
 - **inventory_layers**: FIFO/LIFO costing layers for receipts
 - **inventory**: Current balance by lot code (fast dashboard totals)
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Architecture Diagram

PURCHASE ORDER FLOW

PO Confirmed

- > Create GRN
 - > Record transaction in inventory_transactions (receipt)
 - > qty: +qty, txn_type: 'receipt', category: 'RAW'
 - > Create layer in inventory_layers
 - > qty_available: qty, unit_cost, receipt_date
 - > Create/Update inventory record
 - > qty_available: qty, total_value
- > Dashboard shows RAW stock

PRODUCTION FLOW

Start Production Order

- > Consume RAW → Produce WIP
 - > Record transactions in inventory_transactions
 - > Consume: qty: -qty, txn_type: 'consume', category: 'RAW'
 - > Produce: qty: +qty, txn_type: 'produce', category: 'WIP'

```

> Reduce layers in inventory_layers (FIFO)
  > Deduct from oldest layers, qty_available decreases

> Update inventory records
  > RAW: qty_available decreases
  > WIP: qty_available increases

> Complete Production Items

  > Record transactions in inventory_transactions
    > Consume: qty: -qty, txn_type: 'consume', category: 'WIP'
    > Produce: qty: +qty, txn_type: 'receipt', category: 'FG'

  > Reduce/create layers in inventory_layers
    > Reduce WIP layers (FIFO)
    > Create FG layer with cost

  > Update inventory records
    > WIP: qty_available decreases
    > FG: qty_available increases

> Dashboard shows updated RAW/WIP/FG balances

```

DELIVERY FLOW

Ship Finished Goods to Customer

```

> Record transaction in inventory_transactions (delivery)
  > qty: -qty, txn_type: 'delivery', category: 'FG'

> Reduce layers in inventory_layers (FIFO)
  > Deduct from oldest FG layers

> Update inventory record
  > FG: qty_available decreases

```

Complete Example with Data

Scenario

- Purchase 1,000 units of Raw Material at \$10.00 per unit
- Receive 600 units first (GRN #1), then 400 units (GRN #2) at \$12.00

each

- Start production: consume 750 RAW units → produce 750 WIP
- Complete production: convert 500 WIP → 500 FG
- Deliver 200 FG units to customer

Step 1: Receive First GRN (600 units @ \$10.00)

What happens: - 600 units of RAW material received

Data inserted:

inventory_transactions	lot_code	item_code	category	txn_type	qty	unit_cost	total_cost	warehouse	related_doc			
										RAW-1	RM-001	RAW
receipt	600	10.00	6,000.00	MAIN	GRN-001							

inventory_layers (FIFO Layer A)	lot_code	item_code	category	qty_available	unit_cost	total_cost	receipt_date					
								RAW-1	RM-001	RAW	600	
	10.00	6,000.00	2025-01-15									

inventory	lot_code	item_code	category	qty_available	unit_cost	total_value						
							RAW-1					
	RM-001	RAW	600	10.00	6,000.00							

Dashboard: RAW Materials = 600

Step 2: Receive Second GRN (400 units @ \$12.00)

What happens: - 400 additional RAW units received at higher price

Data inserted:

inventory_transactions	lot_code	item_code	category	txn_type	qty	unit_cost	total_cost	warehouse	related_doc			
										RAW-2	RM-001	RAW
receipt	400	12.00	4,800.00	MAIN	GRN-002							

inventory_layers (FIFO Layer B added)	lot_code	item_code	category	qty_available	unit_cost	total_cost	receipt_date					
								RAW-1	RM-001	RAW	600	
	10.00	6,000.00	2025-01-15					RAW-2	RM-001	RAW	400	
					12.00	4,800.00	2025-01-20					

inventory	lot_code	item_code	category	qty_available	unit_cost	total_value						
							RAW-1					
	RM-001	RAW	600	10.00	6,000.00			RAW-2	RM-001	RAW	400	
					12.00	4,800.00						

Dashboard: RAW Materials = 1,000 (600 + 400)

Step 3: Start Production (Consume 750 RAW → Produce 750 WIP)

What happens: - Production order consumes 750 RAW units - Creates 750 WIP units - FIFO costing: uses oldest layers first

FIFO Layer Consumption: - Layer A (600 @ \$10): fully consumed → cost = \$6,000.00 - Layer B (400 @ \$12): 150 units consumed → cost = \$1,800.00 -
Total cost allocated: \$7,800.00

Data inserted/updated:

inventory_transactions	lot_code	item_code	category	txn_type	qty	unit_cost	total_cost	warehouse	related_doc
consume	-600	10.00	-6,000.00	MAIN	PROD-001	RAW-1	RM-001	RAW	
RAW	consume	-150	12.00	-1,800.00	MAIN	PROD-001	WIP-1	RM-001	
WIP	produce	+750	10.40	7,800.00	PRODUCTION	PROD-001			

inventory_layers (updated)	lot_code	item_code	category	qty_available	unit_cost	total_cost	receipt_date
RAW-1	RM-001	RAW	0	10.00	0.00		
2025-01-15	RAW-2	RM-001	RAW	250	12.00	3,000.00	2025-01-20
WIP-1	RM-001	WIP	750	10.40	7,800.00		2025-01-25

inventory (updated)	lot_code	item_code	category	qty_available	unit_cost	total_value
RAW-2	RM-001	RAW	250	12.00	3,000.00	
WIP-1	RM-001	WIP	750	10.40	7,800.00	

Dashboard: RAW = 250 | WIP = 750 | FG = 0

Step 4: Complete Production Items (500 FG)

What happens: - 500 WIP units converted to 500 FG units - FIFO costing applied to WIP consumption

FIFO WIP Consumption: - From WIP-1 layer (750 @ \$10.40): consume 500 → cost = \$5,200.00

Data inserted/updated:

inventory_transactions	lot_code	item_code	category	txn_type	qty	unit_cost	total_cost	warehouse	related_doc
WIP-1	RM-001	WIP							

```
consume | -500 | 10.40 | -5,200.00 | PRODUCTION | PROD-001 | | FG-1 | BOX-001 | FG | receipt | +500 | 10.40 | 5,200.00 | FINISHED_GOODS | GRN-003 |
```

```
inventory_layers (updated) | lot_code | item_code | category | qty_available | unit_cost | total_cost | receipt_date | |-----|-----|-----|-----|-----|
|-----|-----|-----| RAW-2 | RM-001 | RAW | 250 | 12.00 | 3,000.00 |
2025-01-20 | | WIP-1 | RM-001 | WIP | 250 | 10.40 | 2,600.00 | 2025-01-25 | |
FG-1 | BOX-001 | FG | 500 | 10.40 | 5,200.00 | 2025-01-28 |
```

```
inventory (updated) | lot_code | item_code | category | qty_available | unit_cost | total_value | |-----|-----|-----|-----|-----|
|-----|-----|-----| RAW-2 | RM-001 | RAW | 250 | 12.00 | 3,000.00 | | WIP-1 | RM-001 |
WIP | 250 | 10.40 | 2,600.00 | | FG-1 | BOX-001 | FG | 500 | 10.40 | 5,200.00 |
```

Dashboard: RAW = 250 | WIP = 250 | FG = 500

Step 5: Deliver 200 FG to Customer

What happens: - Ship 200 FG units to customer - FIFO costing applied to delivery

FIFO FG Consumption: - From FG-1 layer (500 @ \$10.40): consume 200 → cost = \$2,080.00

Data inserted/updated:

```
inventory_transactions | lot_code | item_code | category | txn_type | qty | unit_cost | total_cost | warehouse | related_doc | |-----|-----|-----|-----|-----|
|-----|-----|-----| FG-1 | BOX-001 | FG | delivery | -200 | 10.40 | -2,080.00 | FINISHED_GOODS | DELIV-001 |
```

```
inventory_layers (updated) | lot_code | item_code | category | qty_available | unit_cost | total_cost | receipt_date | |-----|-----|-----|-----|-----|
|-----|-----|-----| RAW-2 | RM-001 | RAW | 250 | 12.00 | 3,000.00 |
2025-01-20 | | WIP-1 | RM-001 | WIP | 250 | 10.40 | 2,600.00 | 2025-01-25 | |
FG-1 | BOX-001 | FG | 300 | 10.40 | 3,120.00 | 2025-01-28 |
```

```
inventory (updated) | lot_code | item_code | category | qty_available | unit_cost | total_value | |-----|-----|-----|-----|-----|
|-----|-----|-----| RAW-2 | RM-001 | RAW | 250 | 12.00 | 3,000.00 | | WIP-1 | RM-001 |
WIP | 250 | 10.40 | 2,600.00 | | FG-1 | BOX-001 | FG | 300 | 10.40 | 3,120.00 |
```

Dashboard: RAW = 250 | WIP = 250 | FG = 300

Summary Table

Final State Summary

Transaction Type	RAW	WIP	FG	Total Cost
Receipts	1,000	-	500	\$15,800.00
Consumed	750	500	200	\$13,080.00
Delivered	-	-	200	\$2,080.00
Remaining	250	250	300	\$8,720.00

Key Principles

1. Transaction Flow

Every movement creates an entry in `inventory_transactions` that is never modified.

2. Costing Method (FIFO)

`inventory_layers` keeps receipts by date. Consumes oldest first; updates `qty_available`; deletes empty layers.

3. Current Balance

`inventory` provides per-lot totals for dashboards and reports.

4. Triple Entry Accounting

All three tables stay in sync: - `inventory_transactions` = complete history - `inventory_layers` = costing layers - `inventory` = current balance

How Dashboard Calculations Work

```
-- RAW Materials Total
SELECT SUM(qty_available)
FROM inventory
WHERE category = 'RAW';

-- WIP Total
SELECT SUM(qty_available)
FROM inventory
WHERE category = 'WIP';
```

```

-- Finished Goods Total
SELECT SUM(qty_available)
FROM inventory
WHERE category = 'FG';

-- Total Inventory Value
SELECT SUM(total_value)
FROM inventory;

-- By Warehouse
SELECT warehouse, SUM(qty_available), SUM(total_value)
FROM inventory
GROUP BY warehouse;

```

Common Operations

Add Stock (Receipt)

1. Insert `inventory_transactions` (receipt)
2. Insert `inventory_layers`
3. Insert/update `inventory`

Move Stock (Consume/Produce)

1. Insert `inventory_transactions` (consume)
2. Insert `inventory_transactions` (produce)
3. Reduce layers in `inventory_layers` (FIFO)
4. Create new layers if producing
5. Update both lots in `inventory`

Deliver Stock (Delivery)

1. Insert `inventory_transactions` (delivery)
 2. Reduce layers in `inventory_layers` (FIFO)
 3. Update `inventory`
-

Error Prevention

1. Never delete from `inventory_transactions`
2. Keep layers FIFO sorted by `receipt_date`
3. Update all three tables in a transaction
4. Validate `qty_available >= 0` before consumes
5. Reconcile `inventory` against `inventory_layers`